Curvature computed using exterior differential forms

GRAVITATION (MTW)
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Box 14.5 CURVATURE COMPUTED USING EXTERIOR DIFFERENTIAL
FORMS (METRIC FOR FRIEDMANN COSMOLOGY), Page 355

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(%i2) `info:build_info()$info@version;`

(%o2)

5.38.1

(%i2) `reset()$kill(all)$`

(%i1) `derivabbrev:true$`

(%i2) `ratprint:false$`

(%i3) `fpprintprec:5$`

(%i4) `load(linearalgebra)$`

(%i5) `if get('draw,'version)=false then load(draw)$`

(%i6) `wxplot_size:[1024,768]$`

(%i7) `declare(trigsimp,evfun)$`

(%i8) `if get('itensor,'version)=false then load(itensor)$`

(%i9) `imetric(g)$`

(%i10) `if get('ctensor,'version)=false then load(ctensor)$`

(%i16)
```
ctrgsimp:true$
ratchristof:true$
ratriemann:true$
rateinstein:true$
ratweyl:true$
ratfac:true$
```

(%i17) `if get('cartan,'version)=false then load(cartan)$`

(%i18) `if get('format,'version)=false then load(format)$`

The Friedmann metric

(%i19) ξ`:ct_coords:[t,`χ`,`θ`,`ϕ`]$`

(%i20) `dim:length(ct_coords)$`

(%i21) `depends(a,t)$`

Line Element

(Box 27.1) represents a spacetime where each constant-t hypersurface is a three-dimensional hypersphere of proper circumference $2\pi\, a(t)$. An orthonormal basis is easily found in this spacetime; thus,

(%i22)
```
ldisplay(ds^2=line_element:-del(t)^2+a^2*(del(χ)^2+sin(χ)^2*(del(θ)^2+sin(θ)^2*del(ϕ)^2)))$
```

$$ds^2 = a^2 \left(\mathrm{del}\,(\chi)^2 + \sin(\chi)^2 \left(\sin(\theta)^2\, \mathrm{del}\,(\phi)^2 + \mathrm{del}\,(\theta)^2\right)\right) - \mathrm{del}(t)^2 \tag{\%t22}$$

(%i26)
```
lg:zeromatrix(dim,dim)$
for i thru dim do
lg[i,i]:expand(coeff(expand(line_element),del(ξ[i])^2))$
for j thru dim do for k thru dim do
if j≠k then lg[j,k]:expand(ratsimp(1/2*coeff(coeff(expand(line_element),del(ξ[j])),del(ξ[k]))))$
ishow('g([μ,ν])=lg)$
```

$$\mathrm{g}_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & a^2 & 0 & 0 \\ 0 & 0 & a^2 \sin(\chi)^2 & 0 \\ 0 & 0 & 0 & a^2 \sin(\theta)^2 \sin(\chi)^2 \end{pmatrix} \tag{\%t26}$$

(%i27)
```
cmetric()$
```

(%i28)
```
ishow('g([],[μ,ν])=ug)$
```

$$\mathrm{g}^{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{1}{a^2} & 0 & 0 \\ 0 & 0 & \frac{1}{a^2 \sin(\chi)^2} & 0 \\ 0 & 0 & 0 & \frac{1}{a^2 \sin(\theta)^2 \sin(\chi)^2} \end{pmatrix} \tag{\%t28}$$

Christoffel Symbol of the first kind

(%i30)
```
christof(false)$
for i thru dim do for j:i thru dim do for k thru dim do
if lcs[i,j,k]≠0 then
ishow('Γ([ξ[i],ξ[j],ξ[k]],[])=lcs[i,j,k])$
```

$$\Gamma_{t\chi\chi} = a\ (a_t) \tag{\%t30}$$

$$\Gamma_{t\theta\theta} = a\ (a_t)\ \sin(\chi)^2 \tag{\%t30}$$

$$\Gamma_{t\phi\phi} = a\ (a_t)\ \sin(\theta)^2 \sin(\chi)^2 \tag{\%t30}$$

$$\Gamma_{\chi\chi t} = -a\ (a_t) \tag{\%t30}$$

$$\Gamma_{\chi\theta\theta} = a^2\ \cos(\chi)\ \sin(\chi) \tag{\%t30}$$

$$\Gamma_{\chi\phi\phi} = a^2 \sin(\theta)^2\ \cos(\chi)\ \sin(\chi) \tag{\%t30}$$

$$\Gamma_{\theta\theta t} = -a\ (a_t)\ \sin(\chi)^2 \tag{\%t30}$$

$$\Gamma_{\theta\theta\chi} = -a^2\ \cos(\chi)\ \sin(\chi) \tag{\%t30}$$

$$\Gamma_{\theta\phi\phi} = a^2\ \cos(\theta)\ \sin(\theta) \sin(\chi)^2 \tag{\%t30}$$

$$\Gamma_{\phi\phi t} = -a\ (a_t)\ \sin(\theta)^2 \sin(\chi)^2 \tag{\%t30}$$

$$\Gamma_{\phi\phi\chi} = -a^2 \sin(\theta)^2\ \cos(\chi)\ \sin(\chi) \tag{\%t30}$$

$$\Gamma_{\phi\phi\theta} = -a^2\ \cos(\theta)\ \sin(\theta) \sin(\chi)^2 \tag{\%t30}$$

Christoffel Symbol of the second kind

(%i32)
```
christof(false)$
for i thru dim do for j:i thru dim do for k thru dim do
if mcs[i,j,k]≠0 then
ishow('Γ([ξ[i],ξ[j]],[ξ[k]])=mcs[i,j,k])$
```

$$\Gamma^{\chi}_{t\chi} = \frac{a_t}{a} \tag{\%t32}$$

$$\Gamma^{\theta}_{t\theta} = \frac{a_t}{a} \tag{\%t32}$$

$$\Gamma^{\phi}_{t\phi} = \frac{a_t}{a} \tag{\%t32}$$

$$\Gamma^{t}_{\chi\chi} = a\ (a_t) \tag{\%t32}$$

$$\Gamma^{\theta}_{\chi\theta} = \frac{\cos(\chi)}{\sin(\chi)} \tag{\%t32}$$

$$\Gamma^{\phi}_{\chi\phi} = \frac{\cos(\chi)}{\sin(\chi)} \tag{\%t32}$$

$$\Gamma^{t}_{\theta\theta} = a\ (a_t)\ \sin(\chi)^2 \tag{\%t32}$$

$$\Gamma^{\chi}_{\theta\theta} = -\cos(\chi)\ \sin(\chi) \tag{\%t32}$$

$$\Gamma^{\phi}_{\theta\phi} = \frac{\cos(\theta)}{\sin(\theta)} \tag{\%t32}$$

$$\Gamma^{t}_{\phi\phi} = a\ (a_t)\ \sin(\theta)^2 \sin(\chi)^2 \tag{\%t32}$$

$$\Gamma^{\chi}_{\phi\phi} = -\sin(\theta)^2\ \cos(\chi)\ \sin(\chi) \tag{\%t32}$$

$$\Gamma^{\theta}_{\phi\phi} = -\cos(\theta)\ \sin(\theta) \tag{\%t32}$$

Riemann Tensor

(%i34)
```
riemann(false)$
for a thru dim do for b thru dim do
for c thru (if symmetricp(lg,dim) then b else dim) do
for d thru (if symmetricp(lg,dim) then a else dim) do
if riem[a,b,c,d]≠0 then
ishow('R([" ",ξ[a],ξ[b],ξ[c]],[ξ[d]])=riem[a,b,c,d])$
```

$$\mathbf{R}^{t}_{\chi\chi t} = a\ (a_{tt}) \tag{\%t34}$$

$$\mathbf{R}^{t}_{\theta\theta t} = a\ (a_{tt})\ \sin(\chi)^2 \tag{\%t34}$$

$$\mathbf{R}^{\chi}_{\theta\theta\chi} = \left((a_t)^2 + 1\right)\ \sin(\chi)^2 \tag{\%t34}$$

$$\mathbf{R}^{t}_{\phi\phi t} = a\ (a_{tt})\ \sin(\theta)^2 \sin(\chi)^2 \tag{\%t34}$$

$$\mathbf{R}^{\chi}_{\phi\phi\chi} = \left((a_t)^2 + 1\right) \sin(\theta)^2 \sin(\chi)^2 \tag{\%t34}$$

$$\mathbf{R}^{\theta}_{\phi\phi\theta} = \sin(\theta)^2 \left((a_t)^2 \sin(\chi)^2 - \cos(\chi)^2 + 1\right) \tag{\%t34}$$

(%i36)
```
lriemann(false)$
for a thru dim do for b thru dim do
for c thru (if symmetricp(lg,dim) then b else dim) do
for d thru (if symmetricp(lg,dim) then a else dim) do
if lriem[a,b,c,d]≠0 then
ishow('R([ξ[d],ξ[a],ξ[b],ξ[c]],[])=lriem[a,b,c,d])$
```

$$\mathbf{R}_{t\chi\chi t} = -a\,(a_{tt}) \tag{\%t36}$$

$$\mathbf{R}_{t\theta\theta t} = -a\,(a_{tt}) \sin(\chi)^2 \tag{\%t36}$$

$$\mathbf{R}_{\chi\theta\theta\chi} = a^2 \left((a_t)^2 + 1\right) \sin(\chi)^2 \tag{\%t36}$$

$$\mathbf{R}_{t\phi\phi t} = -a\,(a_{tt}) \sin(\theta)^2 \sin(\chi)^2 \tag{\%t36}$$

$$\mathbf{R}_{\chi\phi\phi\chi} = a^2 \left((a_t)^2 + 1\right) \sin(\theta)^2 \sin(\chi)^2 \tag{\%t36}$$

$$\mathbf{R}_{\theta\phi\phi\theta} = a^2 \sin(\theta)^2 \sin(\chi)^2 \left((a_t)^2 \sin(\chi)^2 - \cos(\chi)^2 + 1\right) \tag{\%t36}$$

(%i38)
```
uriemann(false)$
for a thru dim do for b thru dim do
for c thru (if symmetricp(lg,dim) then b else dim) do
for d thru (if symmetricp(lg,dim) then a else dim) do
if lriem[a,b,c,d]≠0 then
ishow('R([],[ξ[d],ξ[a],ξ[b],ξ[c]])=uriem[a,b,c,d])$
```

$$\mathbf{R}^{t\chi\chi t} = -\frac{a_{tt}}{a^3} \tag{\%t38}$$

$$\mathbf{R}^{t\theta\theta t} = -\frac{a_{tt}}{a^3 \sin(\chi)^2} \tag{\%t38}$$

$$\mathbf{R}^{\chi\theta\theta\chi} = \frac{(a_t)^2 + 1}{a^6 \sin(\chi)^2} \tag{\%t38}$$

$$\mathbf{R}^{t\phi\phi t} = -\frac{a_{tt}}{a^3 \sin(\theta)^2 \sin(\chi)^2} \tag{\%t38}$$

$$\mathbf{R}^{\chi\phi\phi\chi} = \frac{(a_t)^2 + 1}{a^6 \sin(\theta)^2 \sin(\chi)^2} \tag{\%t38}$$

$$\mathbf{R}^{\theta\phi\phi\theta} = \frac{(a_t)^2 \sin(\chi)^2 - \cos(\chi)^2 + 1}{a^6 \sin(\theta)^2 \sin(\chi)^6} \tag{\%t38}$$

Ricci Tensor

(%i41)
```
ric:zeromatrix(dim,dim)$
ricci(false)$
for i thru dim do for j:i thru dim do
if ric[i,j]≠0 then
ishow('R([ξ[i],ξ[j]])=factor(ric[i,j]))$
```

$$\mathbf{R}_{tt} = -\frac{3\,(a_{tt})}{a} \tag{\%t41}$$

$$\mathbf{R}_{\chi\chi} = a\ (a_{tt}) + 2(a_t)^2 + 2 \tag{\%t41}$$

$$\mathbf{R}_{\theta\theta} = a\ (a_{tt})\ \sin(\chi)^2 + 2(a_t)^2 \sin(\chi)^2 + \sin(\chi)^2 - \cos(\chi)^2 + 1 \tag{\%t41}$$

$$\mathbf{R}_{\phi\phi} = \sin(\theta)^2 \left(a\ (a_{tt})\ \sin(\chi)^2 + 2(a_t)^2 \sin(\chi)^2 + \sin(\chi)^2 - \cos(\chi)^2 + 1\right) \tag{\%t41}$$

(%i42)
```
map(ldisp,efe:findde(ric,2))$
```

$$a_{tt} \tag{\%t42}$$

$$a\ (a_{tt}) + 2(a_t)^2 + 2 \tag{\%t43}$$

$$a\ (a_{tt})\ \sin(\chi)^2 + 2(a_t)^2 \sin(\chi)^2 + \sin(\chi)^2 - \cos(\chi)^2 + 1 \tag{\%t44}$$

(%i47)
```
uric:zeromatrix(dim,dim)$
uricci(false)$
for i thru dim do for j:i thru dim do
if uric[i,j]≠0 then
ishow('R([],[ξ[i],ξ[j]])=factor(uric[i,j]))$
```

$$\mathbf{R}^{tt} = \frac{3\,(a_{tt})}{a} \tag{\%t47}$$

$$\mathbf{R}^{\chi\chi} = \frac{a\ (a_{tt}) + 2(a_t)^2 + 2}{a^2} \tag{\%t47}$$

$$\mathbf{R}^{\theta\theta} = \frac{a\ (a_{tt})\ \sin(\chi)^2 + 2(a_t)^2 \sin(\chi)^2 + \sin(\chi)^2 - \cos(\chi)^2 + 1}{a^2 \sin(\chi)^2} \tag{\%t47}$$

$$\mathbf{R}^{\phi\phi} = \frac{a\ (a_{tt})\ \sin(\chi)^2 + 2(a_t)^2 \sin(\chi)^2 + \sin(\chi)^2 - \cos(\chi)^2 + 1}{a^2 \sin(\chi)^2} \tag{\%t47}$$

Scalar curvature

(%i48)
```
factor(radcan(scurvature()));
```

$$\frac{2\left(3a\ (a_{tt})\ \sin(\chi)^2 + 3(a_t)^2 \sin(\chi)^2 + 2\sin(\chi)^2 - \cos(\chi)^2 + 1\right)}{a^2 \sin(\chi)^2} \tag{\%o48}$$

Kretschmann invariant

(%i49) `factor(radcan(rinvariant()));`

$$(4(3a^2\,(a_{tt})^2\sin(\chi)^4+3(a_t)^4\sin(\chi)^4+4(a_t)^2\sin(\chi)^4+2\sin(\chi)^4-2(a_t)^2\cos(\chi)^2\sin(\chi)^2+2(a_t)^2\sin(\chi)^2+\cos(\chi)^4-2\cos(\chi)^2+1) \tag{\%o49}$$

Einstein Tensor

(%i52)
```
ein:zeromatrix(dim,dim)$
einstein(false)$
for i thru dim do for j:i thru dim do
if ein[i,j]≠0 then
ishow('G([ξ[i]],[ξ[j]])=ein[i,j])$
```

$$\mathbf{G}^t_t = -\frac{3(a_t)^2\sin(\chi)^2+2\sin(\chi)^2-\cos(\chi)^2+1}{a^2\sin(\chi)^2} \tag{\%t52}$$

$$\mathbf{G}^\chi_\chi = -\frac{2a\,(a_{tt})\,\sin(\chi)^2+(a_t)^2\sin(\chi)^2-\cos(\chi)^2+1}{a^2\sin(\chi)^2} \tag{\%t52}$$

$$\mathbf{G}^\theta_\theta = -\frac{2a\,(a_{tt})+(a_t)^2+1}{a^2} \tag{\%t52}$$

$$\mathbf{G}^\phi_\phi = -\frac{2a\,(a_{tt})+(a_t)^2+1}{a^2} \tag{\%t52}$$

(%i53) `map(ldisp,efe:findde(ein,2))$`

$$3(a_t)^2\sin(\chi)^2+2\sin(\chi)^2-\cos(\chi)^2+1 \tag{\%t53}$$

$$2a\,(a_{tt})\,\sin(\chi)^2+(a_t)^2\sin(\chi)^2-\cos(\chi)^2+1 \tag{\%t54}$$

$$2a\,(a_{tt})+(a_t)^2+1 \tag{\%t55}$$

(%i58)
```
lein:zeromatrix(dim,dim)$
leinstein(false)$
for i thru dim do for j:i thru dim do
if lein[i,j]≠0 then
ishow('G([ξ[i]],[ξ[j]])=lein[i,j])$
```

$$\mathbf{G}^t_t = \frac{3(a_t)^2\sin(\chi)^2+2\sin(\chi)^2-\cos(\chi)^2+1}{a^2\sin(\chi)^2} \tag{\%t58}$$

$$\mathbf{G}^\chi_\chi = -\frac{2a\,(a_{tt})\,\sin(\chi)^2+(a_t)^2\sin(\chi)^2-\cos(\chi)^2+1}{\sin(\chi)^2} \tag{\%t58}$$

$$\mathbf{G}^\theta_\theta = -\left(2a\,(a_{tt})+(a_t)^2+1\right)\sin(\chi)^2 \tag{\%t58}$$

$$\mathbf{G}^\phi_\phi = -\left(2a\,(a_{tt})+(a_t)^2+1\right)\sin(\theta)^2\sin(\chi)^2 \tag{\%t58}$$

Geodesics

(%i59) `cgeodesic(false)$`

Solve for second derivative of coordinates

(%i60) `geodsol:linsolve(listarray(geod),diff(ξ,s,2))$`

(%i61) `map(ldisp,expand(trigsimp(geodsol)))$`

$$t_{ss} = -a\ (a_t)\ (\chi_s)^2 - a\ (a_t)\ \sin(\theta)^2\ (\phi_s)^2 \sin(\chi)^2 - a\ (a_t)\ (\theta_s)^2 \sin(\chi)^2 \tag{\%t61}$$

$$\chi_{ss} = -\frac{2\,(a_t)\ (t_s)\ (\chi_s)}{a} + \sin(\theta)^2\ (\phi_s)^2\ \cos(\chi)\ \sin(\chi) + (\theta_s)^2\ \cos(\chi)\ \sin(\chi) \tag{\%t62}$$

$$\theta_{ss} = -\frac{2\,(\theta_s)\ \cos(\chi)\ (\chi_s)}{\sin(\chi)} + \cos(\theta)\ \sin(\theta)\ (\phi_s)^2 - \frac{2\,(a_t)\ (t_s)\ (\theta_s)}{a} \tag{\%t63}$$

$$\phi_{ss} = -\frac{2\,(\phi_s)\ \cos(\chi)\ (\chi_s)}{\sin(\chi)} - \frac{2\cos(\theta)\ (\theta_s)\ (\phi_s)}{\sin(\theta)} - \frac{2\,(a_t)\ (t_s)\ (\phi_s)}{a} \tag{\%t64}$$

Initialize cartan package

(%i65) `kill(labels)$`

(%i1) `init_cartan(ξ)$`

(%i2) `ldisplay(cartan_basis,cartan_coords,cartan_dim,extdim)$`

$$cartan_basis = [dt, d\chi, d\theta, d\phi] \tag{\%t2}$$

$$cartan_coords = [t, \chi, \theta, \phi] \tag{\%t3}$$

$$cartan_dim = 4 \tag{\%t4}$$

$$extdim = 4 \tag{\%t5}$$

Coframe σ

(%i6) `σ:[dt,a*dχ,a*sin(χ)*dθ,a*sin(χ)*sin(θ)*dϕ]$`

Exterior derivative of the coframe $\mathrm{d}\sigma$

(%i7) `ldisplay(dσ:ext_diff(σ))$`

$$d\sigma = [0, (a_t)\ dt\ d\chi, (a_t)\ dt\ d\theta\ \sin(\chi) + a\ d\theta\ d\chi\ \cos(\chi), (a_t)\ dt\ d\phi\ \sin(\theta)\ \sin(\chi) + a\ d\theta\ d\phi\ \cos(\theta)\ \sin(\chi) + a\ d\phi\ d\chi\ \sin(\theta)\ \cos(\chi \tag{\%t7}$$

$$0 = \mathrm{d}\omega^{\mu} + \omega^{\mu}{}_{\nu} \wedge \omega^{\nu} \tag{14.31a}$$

$$\mathrm{d}g_{\mu\nu} = \omega_{\mu\nu} + \omega_{\nu\mu} \tag{14.31b}$$

Equation (l4.31b) gives, since $\mathrm{d}g_{\mu\nu} = \mathrm{d}\eta_{\mu\nu} = 0$, just:

$$\omega_{\mu\nu} = -\omega_{\nu\mu}$$

so there are only six 1-forms $\omega_{\mu\nu}$ to be found. Turn to the second basic equation (l4.3la). The game now is to *guess* a solution (because this is so often quicker than using systematic methods) to the equations $0 = \mathrm{d}\omega^{\mu} + \omega^{\mu}{}_{\nu} \wedge \omega^{\nu}$ in which the ω^{ν} and thus also $\mathrm{d}\omega^{\mu}$ are known, and $\omega^{\mu}{}_{\nu}$, are unknown. The solution $\omega^{\mu}{}_{\nu}$ is known to be unique; so guessing (if it leads to any answer) can only give the right answer.

(%i8) `kill(labels,lg,ug)$`

(%i2) `iframe_flag:true$`
`cframe_flag:true$`

(%i4) `lfg:matrix([-1,0,0,0],[0,1,0,0],[0,0,1,0],[0,0,0,1])$`
`fri:apply(diag_matrix,σ/cartan_basis)$`

(%i5) `cmetric()$`

(%i6) `ishow('g([μ,ν])=lg)$`

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & a^2 & 0 & 0 \\ 0 & 0 & a^2 \sin(\chi)^2 & 0 \\ 0 & 0 & 0 & a^2 \sin(\theta)^2 \sin(\chi)^2 \end{pmatrix} \tag{\%t6}$$

(%i7) `ishow('g([],[μ,ν])=ug:invert(lg))$`

$$g^{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{1}{a^2} & 0 & 0 \\ 0 & 0 & \frac{1}{a^2 \sin(\chi)^2} & 0 \\ 0 & 0 & 0 & \frac{1}{a^2 \sin(\theta)^2 \sin(\chi)^2} \end{pmatrix} \tag{\%t7}$$

Christoffel Symbol of the first kind

(%i9)
```
christof(false)$
for i thru dim do for j:i thru dim do for k thru dim do
if lcs[i,j,k]≠0 then
ishow('Γ([ξ[i],ξ[j],ξ[k]],[])=lcs[i,j,k])$
```

$$\Gamma_{\chi\chi t} = -\frac{a_t}{a} \tag{\%t9}$$

$$\Gamma_{\theta\theta t} = -\frac{a_t}{a} \tag{\%t9}$$

$$\Gamma_{\theta\theta\chi} = \frac{\cos(\chi)}{a \sin(\chi)} \tag{\%t9}$$

$$\Gamma_{\phi\phi t} = -\frac{a_t}{a} \tag{\%t9}$$

$$\Gamma_{\phi\phi\chi} = \frac{\cos(\chi)}{a \sin(\chi)} \tag{\%t9}$$

$$\Gamma_{\phi\phi\theta} = \frac{\cos(\theta)}{a \sin(\theta) \sin(\chi)} \tag{\%t9}$$

Christoffel Symbol of the second kind

(%i11)
```
christof(false)$
for i thru dim do for j:i thru dim do for k thru dim do
if mcs[i,j,k]≠0 then
ishow('Γ([ξ[i],ξ[j]],[ξ[k]])=mcs[i,j,k])$
```

$$\Gamma^{t}_{\chi\chi} = \frac{a_t}{a} \tag{\%t11}$$

$$\Gamma^{t}_{\theta\theta} = \frac{a_t}{a} \tag{\%t11}$$

$$\Gamma^{\chi}_{\theta\theta} = \frac{\cos(\chi)}{a \sin(\chi)} \tag{\%t11}$$

$$\Gamma^{t}_{\phi\phi} = \frac{a_t}{a} \tag{\%t11}$$

$$\Gamma^{\chi}_{\phi\phi} = \frac{\cos(\chi)}{a \sin(\chi)} \tag{\%t11}$$

$$\Gamma^{\theta}_{\phi\phi} = \frac{\cos(\theta)}{a \sin(\theta) \sin(\chi)} \tag{\%t11}$$

Riemann Tensor

(%i13)
```
riemann(false)$
for a thru dim do for b thru dim do
for c thru (if symmetricp(lg,dim) then b else dim) do
for d thru (if symmetricp(lg,dim) then a else dim) do
if riem[a,b,c,d]≠0 then
ishow('R([" ",ξ[a],ξ[b],ξ[c]],[ξ[d]])=riem[a,b,c,d])$
```

$$\mathbf{R}^{t}_{\chi\chi t} = a\ (a_{tt}) \tag{\%t13}$$

$$\mathbf{R}^{t}_{\theta\theta t} = a\ (a_{tt})\ \sin(\chi)^2 \tag{\%t13}$$

$$\mathbf{R}^{\chi}_{\theta\theta\chi} = a^2\ \left((a_t)^2 + 1\right)\ \sin(\chi)^2 \tag{\%t13}$$

$$\mathbf{R}^{t}_{\phi\phi t} = a\ (a_{tt})\ \sin(\theta)^2 \sin(\chi)^2 \tag{\%t13}$$

$$\mathbf{R}^{\chi}_{\phi\phi\chi} = a^2\ \left((a_t)^2 + 1\right)\ \sin(\theta)^2 \sin(\chi)^2 \tag{\%t13}$$

$$\mathbf{R}^{\theta}_{\phi\phi\theta} = a^2 \sin(\theta)^2 \sin(\chi)^2\ \left((a_t)^2 \sin(\chi)^2 - \cos(\chi)^2 + 1\right) \tag{\%t13}$$

(%i15)
```
lriemann(false)$
for a thru dim do for b thru dim do
for c thru (if symmetricp(lg,dim) then b else dim) do
for d thru (if symmetricp(lg,dim) then a else dim) do
if lriem[a,b,c,d]≠0 then
ishow('R([ξ[d],ξ[a],ξ[b],ξ[c]],[])=lriem[a,b,c,d])$
```

$$\mathbf{R}_{t\chi\chi t} = -\frac{a_{tt}}{a} \tag{\%t15}$$

$$\mathbf{R}_{t\theta\theta t} = -\frac{a_{tt}}{a} \tag{\%t15}$$

$$\mathbf{R}_{\chi\theta\theta\chi} = \frac{(a_t)^2 + 1}{a^2} \tag{\%t15}$$

$$\mathbf{R}_{t\phi\phi t} = -\frac{a_{tt}}{a} \tag{\%t15}$$

$$\mathbf{R}_{\chi\phi\phi\chi} = \frac{(a_t)^2 + 1}{a^2} \tag{\%t15}$$

$$\mathbf{R}_{\theta\phi\phi\theta} = \frac{(a_t)^2 + 1}{a^2} \tag{\%t15}$$

(%i17)
```
uriemann(false)$
for a thru dim do for b thru dim do
for c thru (if symmetricp(lg,dim) then b else dim) do
for d thru (if symmetricp(lg,dim) then a else dim) do
if lriem[a,b,c,d]≠0 then
ishow('R([],[ξ[d],ξ[a],ξ[b],ξ[c]])=uriem[a,b,c,d])$
```

$$\mathbf{R}^{t\chi\chi t} = -a\ (a_{tt}) \tag{\%t17}$$

$$\mathbf{R}^{t\theta\theta t} = -a\ (a_{tt})\ \sin(\chi)^2 \tag{\%t17}$$

$$\mathbf{R}^{\chi\theta\theta\chi} = a^2 \left(\left(a_t\right)^2 + 1\right) \sin\left(\chi\right)^2 \tag{\%t17}$$

$$\mathbf{R}^{t\phi\phi t} = -a \left(a_{tt}\right) \sin\left(\theta\right)^2 \sin\left(\chi\right)^2 \tag{\%t17}$$

$$\mathbf{R}^{\chi\phi\phi\chi} = a^2 \left(\left(a_t\right)^2 + 1\right) \sin\left(\theta\right)^2 \sin\left(\chi\right)^2 \tag{\%t17}$$

$$\mathbf{R}^{\theta\phi\phi\theta} = a^2 \sin\left(\theta\right)^2 \sin\left(\chi\right)^2 \left(\left(a_t\right)^2 \sin\left(\chi\right)^2 - \cos\left(\chi\right)^2 + 1\right) \tag{\%t17}$$

Ricci Tensor

(%i20)
```
ric:zeromatrix(dim,dim)$
ricci(false)$
for i thru dim do for j:i thru dim do
if ric[i,j]≠0 then
ishow('R([ξ[i],ξ[j]])=factor(ric[i,j]))$
```

$$\mathbf{R}_{tt} = -\frac{3\left(a_{tt}\right)}{a} \tag{\%t20}$$

$$\mathbf{R}_{\chi\chi} = \frac{a\left(a_{tt}\right) + 2\left(a_t\right)^2 + 2}{a^2} \tag{\%t20}$$

$$\mathbf{R}_{\theta\theta} = \frac{a\left(a_{tt}\right) + 2\left(a_t\right)^2 + 2}{a^2} \tag{\%t20}$$

$$\mathbf{R}_{\phi\phi} = \frac{a\left(a_{tt}\right) + 2\left(a_t\right)^2 + 2}{a^2} \tag{\%t20}$$

(%i21)
```
map(ldisp,efe:findde(ric,2))$
```

$$a_{tt} \tag{\%t21}$$

$$a\left(a_{tt}\right) + 2\left(a_t\right)^2 + 2 \tag{\%t22}$$

(%i23)
```
solve(eliminate(efe,[diff(a,t,2)]),diff(a,t));
```

$$\left[a_t = -\%i, a_t = \%i\right] \tag{\%o23}$$

(%i26)
```
uric:zeromatrix(dim,dim)$
uricci(false)$
for i thru dim do for j:i thru dim do
if uric[i,j]≠0 then
ishow('R([],[ξ[i],ξ[j]])=factor(uric[i,j]))$
```

$$\mathbf{R}^{tt} = \frac{3\left(a_{tt}\right)}{a} \tag{\%t26}$$

$$\mathbf{R}^{\chi\chi} = \frac{a\left(a_{tt}\right) + 2\left(a_t\right)^2 + 2}{a^2} \tag{\%t26}$$

$$\mathbf{R}^{\theta\theta} = \frac{a\left(a_{tt}\right) + 2\left(a_t\right)^2 + 2}{a^2} \tag{\%t26}$$

$$\mathbf{R}^{\phi\phi} = \frac{a\left(a_{tt}\right) + 2\left(a_t\right)^2 + 2}{a^2} \tag{\%t26}$$

Scalar curvature

(%i27) `trigsimp(radcan(scurvature()));`

$$\frac{6\left(a\ (a_{tt})+(a_t)^2+1\right)}{a^2} \tag{\%o27}$$

Kretschmann invariant

(%i28) `trigsimp(radcan(rinvariant()));`

$$4(\left((a_t)^4+2(a_t)^2+1\right)\sin(\theta)^2\sin(\chi)^4+\left(\left((a_{tt})^2+(a_t)^4+2(a_t)^2+1\right)\sin(\theta)^2+(a_{tt})^2+(a_t)^4+2(a_t)^2+1\right)\sin(\chi)^2+(a_{tt})^2) \tag{\%o28}$$

Einstein Tensor

(%i31)
```
ein:zeromatrix(dim,dim)$
einstein(false)$
for i thru dim do for j:i thru dim do
if ein[i,j]≠0 then
ishow('G([ξ[i]],[ξ[j]])=ein[i,j])$
```

$$\mathbf{G}_t^t=-\frac{3\left((a_t)^2+1\right)}{a^2} \tag{\%t31}$$

$$\mathbf{G}_\chi^\chi=-\frac{2a\ (a_{tt})+(a_t)^2+1}{a^2} \tag{\%t31}$$

$$\mathbf{G}_\theta^\theta=-\frac{2a\ (a_{tt})+(a_t)^2+1}{a^2} \tag{\%t31}$$

$$\mathbf{G}_\phi^\phi=-\frac{2a\ (a_{tt})+(a_t)^2+1}{a^2} \tag{\%t31}$$

(%i32) `map(ldisp,efe:findde(ein,2))$`

$$(a_t)^2+1 \tag{\%t32}$$

$$2a\ (a_{tt})+(a_t)^2+1 \tag{\%t33}$$

(%i36)
```
lein:zeromatrix(dim,dim)$
leinstein(false)$
for i thru dim do for j:i thru dim do
if lein[i,j]≠0 then
ishow('G([ξ[i]],[ξ[j]])=lein[i,j])$
```

$$\mathbf{G}_t^t=\frac{3\left((a_t)^2+1\right)}{a^2} \tag{\%t36}$$

$$\mathbf{G}_\chi^\chi=-2a\ (a_{tt})-(a_t)^2-1 \tag{\%t36}$$

$$\mathbf{G}_\theta^\theta=-\left(2a\ (a_{tt})+(a_t)^2+1\right)\sin(\chi)^2 \tag{\%t36}$$

$$\mathbf{G}_{\phi}^{\phi} = -\left(2a\ (a_{tt}) + (a_t)^2 + 1\right)\ \sin(\theta)^2 \sin(\chi)^2 \tag{\%t36}$$

Geodesics

(%i37) `cgeodesic(false)$`

Solve for second derivative of coordinates

(%i38) `geodsol:linsolve(listarray(geod),diff(ξ,s,2))$`

(%i39) `map(ldisp,expand(trigsimp(geodsol)))$`

$$t_{ss} = -a\ (a_t)\ (\chi_s)^2 - a\ (a_t)\ \sin(\theta)^2\ (\phi_s)^2 \sin(\chi)^2 - a\ (a_t)\ (\theta_s)^2 \sin(\chi)^2 \tag{\%t39}$$

$$\chi_{ss} = -\frac{2\,(a_t)\ (t_s)\ (\chi_s)}{a} + \sin(\theta)^2\ (\phi_s)^2\ \cos(\chi)\ \sin(\chi) + (\theta_s)^2\ \cos(\chi)\ \sin(\chi) \tag{\%t40}$$

$$\theta_{ss} = -\frac{2\,(\theta_s)\ \cos(\chi)\ (\chi_s)}{\sin(\chi)} + \cos(\theta)\ \sin(\theta)\ (\phi_s)^2 - \frac{2\,(a_t)\ (t_s)\ (\theta_s)}{a} \tag{\%t41}$$

$$\phi_{ss} = -\frac{2\,(\phi_s)\ \cos(\chi)\ (\chi_s)}{\sin(\chi)} - \frac{2\cos(\theta)\ (\theta_s)\ (\phi_s)}{\sin(\theta)} - \frac{2\,(a_t)\ (t_s)\ (\phi_s)}{a} \tag{\%t42}$$

Generic Connection 1-form ω

(%i43)
```
kill(labels,ω,Θ,Ω)$
```

(%i6)
```
A:[a_1,a_2,a_3,a_4]$
B:[b_1,b_2,b_3,b_4]$
C:[c_1,c_2,c_3,c_4]$
D:[d_1,d_2,d_3,d_4]$
E:[e_1,e_2,e_3,e_4]$
F:[f_1,f_2,f_3,f_4]$
```

(%i13)
```
ω:zeromatrix(dim,dim)$ ω[1,2]:-ω[2,1]:A.cartan_basis$
ω[1,3]:-ω[3,1]:B.cartan_basis$
ω[1,4]:-ω[4,1]:C.cartan_basis$
ω[2,3]:-ω[3,2]:D.cartan_basis$
ω[2,4]:-ω[4,2]:E.cartan_basis$
ω[3,4]:-ω[4,3]:F.cartan_basis$
```

$\Theta^a = \omega^a_b \wedge \sigma^b$

(%i14)
```
block([matrix_element_mult:"~"],
Θ:list_matrix_entries(expand(ω.σ)))$
```

Cartan's First structural equation: $\mathrm{d}e^{\hat{\alpha}} + \omega^{\hat{\alpha}}_{\hat{\beta}} \wedge e^{\hat{\beta}} = 0$

(%i15)
```
Eq:zeromatrix(dim,dim)$
```

(%i16)
```
for i thru dim do for j thru dim do
Eq[i,j]:format(coeff(coeff(dσ,cartan_basis[i]),cartan_basis[j])=
coeff(coeff(-Θ,cartan_basis[i]),cartan_basis[j]),%list)$
```

(%i17)
```
Eqs:apply('append,list_matrix_entries(Eq))$
```

(%i18)
```
linsol:linsolve(Eqs,append(A,B,C,D,E,F))$
```

solve: dependent equations eliminated: (1 64 63 62 61 2 3 4 21 22 23 24 41 42 43 44 17 33 38 47 49 30 19 20 45 34 37 36 51

(%i19)
```
ldisplay(Θ:at(Θ,linsol))$
```

$$\Theta = [0, -(a_t)\ dt\ d\chi, -(a_t)\ dt\ d\theta\ \sin(\chi) - a\ d\theta\ d\chi\ \cos(\chi), -(a_t)\ dt\ d\phi\ \sin(\theta)\ \sin(\chi) - a\ d\theta\ d\phi\ \cos(\theta)\ \sin(\chi) - a\ d\phi\ d\chi\ \sin(\theta)$$ (%t19)

(%i20)
```
is(dσ=-Θ);
```

$$\text{true} \qquad (\%o20)$$

Update Connection 1-form ω

(%i21)
```
ldisplay(ω:at(ω,linsol))$
```

$$\omega = \begin{pmatrix} 0 & -(a_t)\ d\chi & -(a_t)\ d\theta\ \sin(\chi) & -(a_t)\ d\phi\ \sin(\theta)\ \sin(\chi) \\ (a_t)\ d\chi & 0 & -d\theta\ \cos(\chi) & -d\phi\ \sin(\theta)\ \cos(\chi) \\ (a_t)\ d\theta\ \sin(\chi) & d\theta\ \cos(\chi) & 0 & -d\phi\ \cos(\theta) \\ (a_t)\ d\phi\ \sin(\theta)\ \sin(\chi) & d\phi\ \sin(\theta)\ \cos(\chi) & d\phi\ \cos(\theta) & 0 \end{pmatrix} \qquad (\%t21)$$

(%i22)
```
for i thru dim do for j:i thru dim do
if ω[i,j]≠0 then
ishow('ω(["  ",ξ[i]],[ξ[j]])=trigsimp(ω[i,j]))$
```

$$\omega_t^{\chi} = -\left(a_t\right)\, d\chi \tag{\%t22}$$

$$\omega_t^{\theta} = -\left(a_t\right)\, d\theta\, \sin\left(\chi\right) \tag{\%t22}$$

$$\omega_t^{\phi} = -\left(a_t\right)\, d\phi\, \sin\left(\theta\right)\, \sin\left(\chi\right) \tag{\%t22}$$

$$\omega_{\chi}^{\theta} = -d\theta\, \cos\left(\chi\right) \tag{\%t22}$$

$$\omega_{\chi}^{\phi} = -d\phi\, \sin\left(\theta\right)\, \cos\left(\chi\right) \tag{\%t22}$$

$$\omega_{\theta}^{\phi} = -d\phi\, \cos\left(\theta\right) \tag{\%t22}$$

Update Connection 2-form $\mathrm{d}\omega$

(%i23)
```
ldisplay(dω:trigsimp(matrixmap(edit,ext_diff(ω))))$
```

$$d\omega = \begin{pmatrix} 0 & -\left(a_{tt}\right)\, dt\, d\chi \\ \left(a_{tt}\right)\, dt\, d\chi & 0 \\ d\theta\, \left(\left(a_{tt}\right)\, dt\, \sin\left(\chi\right) + \left(a_t\right)\, d\chi\, \cos\left(\chi\right)\right) & -d\theta\, d\chi\, \sin\left(\chi\right) \\ d\phi\, \left(\left(a_{tt}\right)\, dt\, \sin\left(\theta\right)\, \sin\left(\chi\right) + \left(a_t\right)\, d\theta\, \cos\left(\theta\right)\, \sin\left(\chi\right) + \left(a_t\right)\, d\chi\, \sin\left(\theta\right)\, \cos\left(\chi\right)\right) & d\phi\, \left(d\theta\, \cos\left(\theta\right)\, \cos\left(\chi\right) - d\chi\, \sin\left(\theta\right)\, \sin\right. \end{pmatrix} \tag{\%t23}$$

Update coefficients

(%i29)
```
ldisplay(A:at(A,linsol))$
ldisplay(B:at(B,linsol))$
ldisplay(C:at(C,linsol))$
ldisplay(D:at(D,linsol))$
ldisplay(E:at(E,linsol))$
ldisplay(F:at(F,linsol))$
```

$$A = \left[0, a_t, 0, 0\right] \tag{\%t24}$$

$$B = \left[0, 0, \left(a_t\right)\, \sin\left(\chi\right), 0\right] \tag{\%t25}$$

$$C = \left[0, 0, 0, \left(a_t\right)\, \sin\left(\theta\right)\, \sin\left(\chi\right)\right] \tag{\%t26}$$

$$D = \left[0, 0, \cos\left(\chi\right), 0\right] \tag{\%t27}$$

$$E = \left[0, 0, 0, \sin\left(\theta\right)\, \cos\left(\chi\right)\right] \tag{\%t28}$$

$$F = \left[0, 0, 0, \cos\left(\theta\right)\right] \tag{\%t29}$$

Cartan's Second structural equation: $R_{\hat{\beta}}^{\hat{\alpha}} = \mathrm{d}\omega_{\hat{\beta}}^{\hat{\alpha}} + \omega_{\hat{\gamma}}^{\hat{\alpha}} \wedge \omega_{\hat{\beta}}^{\hat{\gamma}}$

Curvature 2-form Ω

$$R^a{}_b = \mathrm{d}\omega^a{}_b + \omega^a{}_c \wedge \omega^c{}_b$$

(%i30)
```
block([matrix_element_mult:"~"], ldisplay(Ω:trigsimp(matrixmap(edit,dω+ω.ω))))$
```

$$\Omega = \begin{pmatrix} 0 & -\left(a_{tt}\right)\, dt\, d\chi & -\left(a_{tt}\right)\, dt\, d\theta\, \sin\left(\chi\right) & -\left(a_{tt}\right)\, dt\, d\phi\, \sin \\ \left(a_{tt}\right)\, dt\, d\chi & 0 & -\left(\left(a_t\right)^2 - 1\right)\, d\theta\, d\chi\, \sin\left(\chi\right) & -\left(\left(a_t\right)^2 - 1\right)\, d\phi\, d\chi \\ \left(a_{tt}\right)\, dt\, d\theta\, \sin\left(\chi\right) & \left(\left(a_t\right)^2 - 1\right)\, d\theta\, d\chi\, \sin\left(\chi\right) & 0 & -\left(\left(a_t\right)^2 - 1\right)\, d\theta\, d\phi \\ \left(a_{tt}\right)\, dt\, d\phi\, \sin\left(\theta\right)\, \sin\left(\chi\right) & \left(\left(a_t\right)^2 - 1\right)\, d\phi\, d\chi\, \sin\left(\theta\right)\, \sin\left(\chi\right) & \left(\left(a_t\right)^2 - 1\right)\, d\theta\, d\phi\, \sin\left(\theta\right) \sin\left(\chi\right)^2 & 0 \end{pmatrix} \tag{\%t30}$$

(%i31)
```
for i thru dim do for j:i thru dim do
if ω[i,j]≠0 then
ishow('Ω(["  ",ξ[i]],[ξ[j]])=trigsimp(Ω[i,j]))$
```

$$\Omega_t^{\chi} = -\left(a_{tt}\right)\ dt\ d\chi \tag{\%t31}$$

$$\Omega_t^{\theta} = -\left(a_{tt}\right)\ dt\ d\theta\ \sin\left(\chi\right) \tag{\%t31}$$

$$\Omega_t^{\phi} = -\left(a_{tt}\right)\ dt\ d\phi\ \sin\left(\theta\right)\ \sin\left(\chi\right) \tag{\%t31}$$

$$\Omega_{\chi}^{\theta} = -\left(\left(a_t\right)^2 - 1\right)\ d\theta\ d\chi\ \sin\left(\chi\right) \tag{\%t31}$$

$$\Omega_{\chi}^{\phi} = -\left(\left(a_t\right)^2 - 1\right)\ d\phi\ d\chi\ \sin\left(\theta\right)\ \sin\left(\chi\right) \tag{\%t31}$$

$$\Omega_{\theta}^{\phi} = -\left(\left(a_t\right)^2 - 1\right)\ d\theta\ d\phi\ \sin\left(\theta\right)\sin\left(\chi\right)^2 \tag{\%t31}$$

Forms in terms of the coframe

(%i32)
```
Eqs:makelist(o[ξ[i]]=σ[i],i,1,cartan_dim)$
```

(%i33)
```
Eqs:ishow(makelist(o([],[ξ[i]])=σ[i],i,1,cartan_dim))$
```

$$\left[o^t = dt, o^{\chi} = a\ d\chi, o^{\theta} = a\ d\theta\ \sin\left(\chi\right), o^{\phi} = a\ d\phi\ \sin\left(\theta\right)\ \sin\left(\chi\right)\right] \tag{Eqs}$$

(%i34)
```
linsolc:ishow(linsolve(Eqs,cartan_basis))$
```

$$\left[dt = o^t, d\chi = \frac{o^{\chi}}{a}, d\theta = \frac{o^{\theta}}{a\ \sin\left(\chi\right)}, d\phi = \frac{o^{\phi}}{a\ \sin\left(\theta\right)\ \sin\left(\chi\right)}\right] \tag{linsolc}$$

Connection 1-form ω

(%i35)
```
ishow(ev(ω,linsolc,fullratsimp))$
```

$$\begin{pmatrix} 0 & -\frac{o^{\chi}\ \left(a_t\right)}{a} & -\frac{o^{\theta}\ \left(a_t\right)}{a} & -\frac{o^{\phi}\ \left(a_t\right)}{a} \\ \frac{o^{\chi}\ \left(a_t\right)}{a} & 0 & -\frac{o^{\theta}\ \cos\left(\chi\right)}{a\ \sin\left(\chi\right)} & -\frac{o^{\phi}\ \cos\left(\chi\right)}{a\ \sin\left(\chi\right)} \\ \frac{o^{\theta}\ \left(a_t\right)}{a} & \frac{o^{\theta}\ \cos\left(\chi\right)}{a\ \sin\left(\chi\right)} & 0 & -\frac{o^{\phi}\ \cos\left(\theta\right)}{a\ \sin\left(\theta\right)\ \sin\left(\chi\right)} \\ \frac{o^{\phi}\ \left(a_t\right)}{a} & \frac{o^{\phi}\ \cos\left(\chi\right)}{a\ \sin\left(\chi\right)} & \frac{o^{\phi}\ \cos\left(\theta\right)}{a\ \sin\left(\theta\right)\ \sin\left(\chi\right)} & 0 \end{pmatrix} \tag{\%t35}$$

Curvature 2-form Ω

(%i36)
```
ishow(ev(Ω,linsolc,fullratsimp))$
```

$$\begin{pmatrix} 0 & -\frac{o^t\ o^{\chi}\ \left(a_{tt}\right)}{a} & -\frac{o^t\ o^{\theta}\ \left(a_{tt}\right)}{a} & -\frac{o^t\ o^{\phi}\ \left(a_{tt}\right)}{a} \\ \frac{o^t\ o^{\chi}\ \left(a_{tt}\right)}{a} & 0 & -\frac{o^{\theta}\ o^{\chi}\ \left(\left(a_t\right)^2-1\right)}{a^2} & -\frac{o^{\phi}\ o^{\chi}\ \left(\left(a_t\right)^2-1\right)}{a^2} \\ \frac{o^t\ o^{\theta}\ \left(a_{tt}\right)}{a} & \frac{o^{\theta}\ o^{\chi}\ \left(\left(a_t\right)^2-1\right)}{a^2} & 0 & -\frac{o^{\theta}\ o^{\phi}\ \left(\left(a_t\right)^2-1\right)}{a^2} \\ \frac{o^t\ o^{\phi}\ \left(a_{tt}\right)}{a} & \frac{o^{\phi}\ o^{\chi}\ \left(\left(a_t\right)^2-1\right)}{a^2} & \frac{o^{\theta}\ o^{\phi}\ \left(\left(a_t\right)^2-1\right)}{a^2} & 0 \end{pmatrix} \tag{\%t36}$$

Time

(%i37) `elapsed_real_time();`

$$34.639 \tag{\%o37}$$

(%i38) `elapsed_run_time();`

$$3.406 \tag{\%o38}$$